



AX Series

AX-12A



AX-12+, AX-12A

WARNING

NOTE : The XL430-W250 is an X series DYNAMIXEL serving as a direct replacement of the AX-12. X series servos offer improved performance and additional features, as well as support for DYNAMIXEL Protocol 2.0. ROBOTIS recommends that new and existing AX-12 users transition to XL430-W250.

NOTE

NOTE : AX-12+ is the improved version of existing AX-12; the design of circuit, material, and wheel gear are specially improved.

WARNING

NOTE : AX-12A is a new version of the AX-12+ with the same performance but more advanced external design. Only the AX-12A is now being sold.

Specifications

Item	Specifications
Baud Rate	7,843 [bps] ~ 1 [Mbps]
Weight	AX-12 (53.5 [g]), AX-12+ (53.5 [g]), AX-12A (54.6 [g])

Item	Specifications
Dimensions (W x H x D)	32 X 50 X 40 [mm] 1.26 X 1.97 X 1.57 [inch]
Resolution	0.29 [°]
Running Degree	0 ~ 300 [°] Endless Turn
Motor	Cored
Gear Ratio	254 : 1
Stall Torque	1.5 [N.m] (at 12 [V], 1.5 [A])
No Load Speed	59 [rev/min] (at 12V)
Operating Temperature	-5 ~ +70 [°C]
Input Voltage	9.0 ~ 12.0 [V] (Recommended : 11.1V)
Command Signal	Digital Packet
Physical Connection	TTL Level Multi Drop Bus Half Duplex Asynchronous Serial Communication (8bit, 1stop, No Parity)
ID	254 ID (0~253)
Feedback	Position, Temperature, Load, Input Voltage, etc
Gear Material	Engineering Plastic(Full)
Case Material	Engineering Plastic(Front, Middle, Back)

 NOTE

NOTE : Stall torque is the maximum instantaneous and static torque. Stable motions are possible with robots designed for loads with 1/5 or less of the stall torque.

 DANGER



DANGER (Ignoring these warnings may cause serious injury or death)

- Never place items containing water, flammables/open flames, or solvents near the product.
- Never place fingers, arms, toes, and other body parts near product during operation.
- Cease operation and remove power from the product if the product begins to emit strange odors, noises, or smoke.
- Keep product out of reach of children.
- Check input polarity before installing or energizing wiring or cables.

 WARNING



CAUTION (Ignoring these warnings may cause mild injury or damage to the product)

- Always comply with the product's official operating environment specifications including input voltage, current, and operating temperature.
- Do not insert blades or other sharp objects during product operation.

 WARNING



ATTENTION (Ignoring these warnings may cause minor injury or damage to the product)

- Do not disassemble or modify the product.
- Do not drop the product or apply strong impacts.
- Do not connect or disconnect DYNAMIXEL cables while power is being supplied.

Control Table

The Control Table is a data structure used by DYNAMIXEL actuators to manage the state of the device. Users can read data registers to get information about the status of the device with Read Instruction Packets, and modify data registers to control the device with Write Instruction Packets.

Control Table, Data, Address

The Control Table is a structure that consists of multiple Data fields to store status or to control the device. Users can check current status of the device by reading a specific Data from the Control Table with Read Instruction Packets. WRITE Instruction Packets enable users to control the device by changing specific Data in the Control Table. The Address is a unique value when accessing a specific Data in the Control Table with Instruction Packets. In order to read or write data, users must designate a specific Address in the Instruction Packet. Please refer to [DYNAMIXEL Protocol 1.0](#) for more details about Instruction Packets.

Area (EEPROM, RAM)

The Control Table is divided into 2 Areas. Data in the RAM Area is reset to initial values when the power is reset(Volatile). On the other hand, data in the EEPROM Area is maintained even when the device is powered off(Non-Volatile).

Size

The Size of data varies from 1 ~ 2 bytes depend on their usage. Please check the size of data when updating the data with an Instruction Packet. For data larger than 2 bytes will be saved according to [Little Endian](#).

Access

The Control Table has two different access properties. 'RW' property stands for read and write access permission while 'R' stands for read only access permission. Data with the read only property cannot be changed by the WRITE Instruction. Read only property('R') is generally used for measuring and monitoring purpose, and read write property('RW') is used for controlling device.

Initial Value

Each data in the Control Table is restored to initial values when the device is turned on. Default values in the EEPROM area are initial values of the device (factory default settings). If any values

in the EEPROM area are modified by a user, modified values will be restored as initial values when the device is turned on. Initial Values in the RAM area are restored when the device is turned on.

Control Table of EEPROM Area

Address	Size(Byte)	Data Name	Description	Access	Initial Value
0	2	Model Number	Model Number	R	12
2	1	Firmware Version	Firmware Version	R	-
3	1	ID	DYNAMIXEL ID	RW	1
4	1	Baud Rate	Communication Speed	RW	1
5	1	Return Delay Time	Response Delay Time	RW	250
6	2	CW Angle Limit	Clockwise Angle Limit	RW	0
8	2	CCW Angle Limit	Counter-Clockwise Angle Limit	RW	1023
11	1	Temperature Limit	Maximum Internal Temperature Limit	RW	70
12	1	Min Voltage Limit	Minimum Input Voltage Limit	RW	60
13	1	Max Voltage Limit	Maximum Input Voltage Limit	RW	140
14	2	Max Torque	Maximum Torque	RW	1023
16	1	Status Return Level	Select Types of Status Return	RW	2

Address	Size(Byte)	Data Name	Description	Access	Initial Value
17	1	Alarm LED	LED for Alarm	RW	36
18	1	Shutdown	Shutdown Error Information	RW	36

Control Table of RAM Area

Address	Size(Byte)	Data Name	Description	Access	Initial Value
24	1	Torque Enable	Motor Torque On/Off	RW	0
25	1	LED	Status LED On/Off	RW	0
26	1	CW Compliance Margin	CW Compliance Margin	RW	1
27	1	CCW Compliance Margin	CCW Compliance Margin	RW	1
28	1	CW Compliance Slope	CW Compliance Slope	RW	32
29	1	CCW Compliance Slope	CCW Compliance Slope	RW	32
30	2	Goal Position	Target Position	RW	-
32	2	Moving Speed	Moving Speed	RW	-
34	2	Torque Limit	Torque Limit	RW	Max Torque
36	2	Present Position	Present Position	R	-

Address	Size(Byte)	Data Name	Description	Access	Initial Value
38	2	Present Speed	Present Speed	R	-
40	2	Present Load	Present Load	R	-
42	1	Present Voltage	Present Voltage	R	-
43	1	Present Temperature	Present Temperature	R	-
44	1	Registered	If Instruction is registered	R	0
46	1	Moving	Movement Status	R	0
47	1	Lock	Locking EEPROM	RW	0
48	2	Punch	Minimum Current Threshold	RW	32

Control Table Description

Model Number (0)

This address stores model number of DYNAMIXEL.

Firmware Version (2)

This address stores firmware version of DYNAMIXEL.

ID (3)

The ID is a unique value in the network to identify each DYNAMIXEL with an Instruction Packet. 0~253 (0xFD) values can be used as an ID, and 254(0xFE) is occupied as a broadcast ID. The Broadcast ID(254, 0xFE) can send an Instruction Packet to all connected DYNAMIXEL simultaneously.



Please avoid using an identical ID for multiple DYNAMIXEL. You may face communication failure or may not be able to detect DYNAMIXEL with an identical ID.

i NOTE

If the Instruction Packet ID is set to the Broadcast ID(0xFE), Status Packets will not be returned for READ or WRITE Instructions regardless of the set value of Status Return Level (16). For more details, please refer to the [Status Packet](#) section for [DYNAMIXEL Protocol 1.0](#).

Baud Rate (4)

Baud Rate determines serial communication speed between a controller and DYNAMIXEL's.

Value	Baud Rate(bps)	Margin of Error
1(Default)	1M	0.000%
3	500,000	0.000%
4	400,000	0.000%
7	250,000	0.000%
9	200,000	0.000%
16	115200	-2.124%
34	57600	0.794%
103	19200	-0.160%
207	9600	-0.160%

i NOTE

NOTE : Less than 3% of the baud rate error margin will not affect to UART communication.

i NOTE

NOTE : For the stable communication with higher baudrate, configure USB Latency value to the lower. [USB Latency Setting](#)

Return Delay Time (5)

If the DYNAMIXEL receives an Instruction Packet, it will return the Status Packet after the time of the set Return Delay Time(5). Note that the range of values is 0 to 254 (0xFE) and its unit is 2 [μ sec]. For instance, if the Return Delay Time(5) is set to '10', the Status Packet will be returned after 20[μ sec] when the Instruction Packet is received.

Unit	Value Range	Description
2[μ sec]	0 ~ 254	Default value '250'(500[μ sec]) Maximum value: '508'[μ sec]

CW/CCW Angle Limit(6, 8)

The angle limit allows the motion to be restrained. The range and the unit of the value is the same as [Goal Position\(30\)](#).

- CW Angle Limit: the minimum value of [Goal Position\(30\)](#)
- CCW Angle Limit: the maximum value of [Goal Position\(30\)](#) The following two modes can be set pursuant to the value of CW and CCW.

Operation Type	CW / CCW
Wheel Mode	both are 0
Joint Mode	neither are 0

The wheel mode can be used to wheel-type operation robots since motors of the robots spin infinitely. The joint mode can be used to multi-joints robot since the robots can be controlled with specific angles.

Temperature Limit (11)

The operating temperature's upper bound of the DYNAMIXEL.

For instance, if the value of the Temperature Limit(11) is set as '80', it means 80 °C If the operating DYNAMIXEL's internal temperature is over the Temperature Limit(11), the Status

Packet transmits Overheating Error Bit (Bit2) via the ERROR field. Providing that Overheating flag is set in the [Alarm LED\(17\)/Shutdown\(18\)](#), the Alarm LED will start blinking and the motor's output will be 0 [%].

Unit	Value Range
About 1°C	0 ~ 100

WARNING

CAUTION : Do not set the temperature higher than the default value. When the temperature alarm shutdown occurs, wait 20 minutes to cool the temperature before re-use. Keep using the product when the temperature is high can cause severe damage.

Min/Max Voltage Limit (12, 13)

The operating voltage's upper and lower bound of the DYNAMIXEL.

Unit	Value Range	Description
About 0.1V	50 ~ 160	5.0 ~ 16.0V

The range of value is from 50 to 160 and the per unit value is 0.1 [V]. For instance, if the value of the Max Voltage Limit(13) is set as '80', it means 8 V. If the [Present Voltage\(42\)](#) which is representing the input voltage to the device (the DYNAMIXEL) is lower than the Min Voltage Limit(12) or over the Max Voltage Limit(13), the Status Packet transmits Input Voltage Error Bit(0×01) via the ERROR field. Providing that Input Voltage Error Bit(0×01) flag is set in the [Alarm LED\(17\)/Shutdown\(18\)](#), the Alarm LED will start blinking and the motor's output will be 0 [%].

Max Torque (14)

It is the torque value of maximum output. 0 to 1,023 (0×3FF) can be used, and the unit is about 0.1%. For example, Data 1,023 (0×3FF) means that DYNAMIXEL will use 100% of the maximum torque it can produce while Data 512 (0×200) means that DYNAMIXEL will use 50% of the maximum torque. When the power is turned on, [Torque Limit\(34\)](#) is reset to the value of Max Torque(14).

Status Return Level (16)

The Status Return Level (16) decides how to return Status Packet when DYNAMIXEL receives an Instruction Packet.

Value	Responding Instructions	Description
0	PING Instruction	Returns the Status Packet for PING Instruction only
1	PING Instruction READ Instruction	Returns the Status Packet for PING and READ Instruction
2	All Instructions	Returns the Status Packet for all Instructions

i NOTE

If the Instruction Packet ID is set to the Broadcast ID(0xFE), Status Packet will not be returned for READ or WRITE Instructions regardless of Status Return Level (16). For more details, please refer to the [Status Packet](#) section for [DYNAMIXEL Protocol 1.0](#).

Alarm LED(17), Shutdown(18)

The DYNAMIXEL can protect itself by detecting dangerous situations that could occur during the operation. Each Bit is inclusively processed with the 'OR' logic, therefore, multiple options can be generated. For instance, when '0x05' (binary : 00000101) is defined in Shutdown(18), DYNAMIXEL can detect both Input Voltage Error(binary : 00000001) and Overheating Error(binary : 00000100). If those errors are detected, the Alarm LED will start blinking and the motor's output will be 0 [%]. The followings are detectable situations.

Bit	Item	Description
Bit 7	0	-
Bit 6	Instruction Error	Detects that undefined Instruction is transmitted or the ACTION command is delivered without the REG_WRITE command
Bit 5	Overload Error	Detects that persistent load exceeds maximum output

Bit	Item	Description
Bit 4	Checksum Error	Detects that the Checksum of the transmitted Instruction Packet is invalid
Bit 3	Range Error	Detects that the command is given beyond the range of usage
Bit 2	Overheating Error	Detects that the internal temperature exceeds the set temperature
Bit 1	Angle Limit Error	Detects that Goal Position is written with the value that is not between CW Angle Limit and CCW Angle Limit
Bit 0	Input Voltage Error	Detects that input voltage exceeds the configured operating voltage

i NOTE

If Shutdown occurs, **LED will flicker every second.**

Torque Enable (24)

Value	Description
0(Default)	Torque Off
1	Torque On

LED (25)

The LED(65) determines LED On or Off.

Bit	Description
0(Default)	Turn OFF the LED
1	Turn ON the LED

i NOTE

The LED is also used to indicate various statuses of the DYNAMIXEL actuator, refer to the following chart for more information.

Status	LED Representation
Booting	LED blinks once
Factory Reset	LED blinks quickly 4 times
Shutdown Error	LED blinks continuously
Bootloader Mode	LED on continuously

Compliance Margin (26, 27)

It exists in each direction of CW/CCW and means the error between goal position and present position. The range of the value is 0~255, and the unit is the same as Goal Position.(Address 30,31) The greater the value, the more difference occurs.

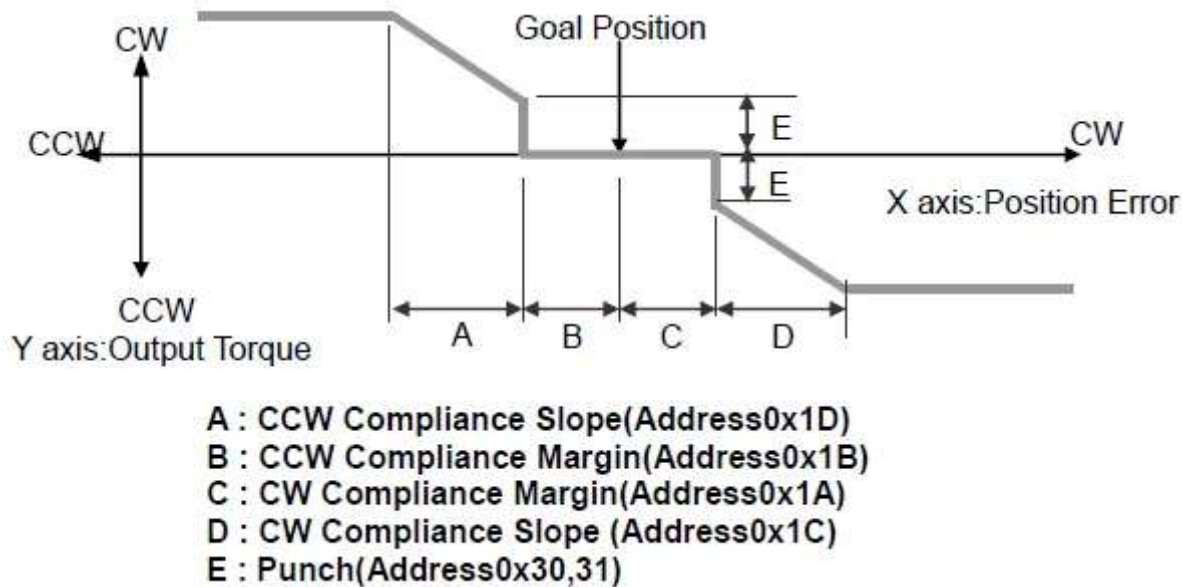
Compliance Slope (28, 29)

It exists in each direction of CW/CCW and sets the level of Torque near the goal position. Compliance Slope is set in 7 steps, the higher the value, the more flexibility is obtained. Data representative value is actually used value. That is, even if the value is set to 25, 16 is used internally as the representative value.

Step	Data Value	Data Representative Value
1	0(0×00) ~ 3(0×03)	2(0×02)
2	4(0×04) ~ 7(0×07)	4(0×04)
3	8(0×08)~15(0×0F)	8(0×08)
4	16(0×10)~31(0×1F)	16(0×10)
5	32(0×20)~63(0×3F)	32(0×20)

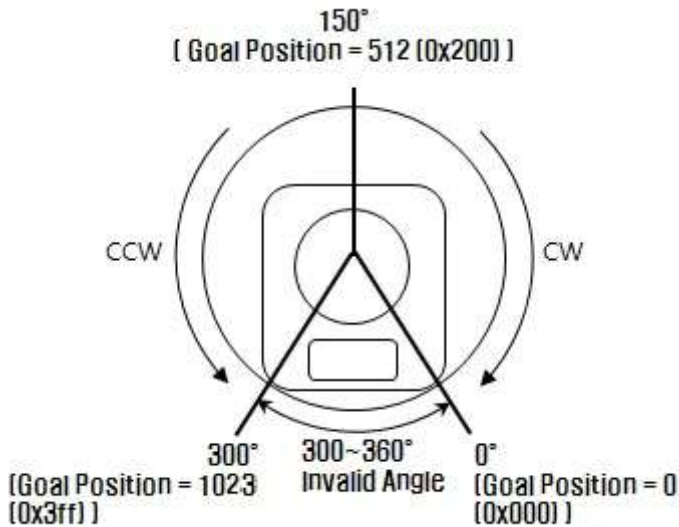
Step	Data Value	Data Representative Value
6	64(0×40)~127(0×7F)	64(0×40)
7	128(0×80)~254(0xFE)	128(0×80)

Compliance is to set the control flexibility of the motor. The following diagram shows the relationship between output torque and position error of the motor.



Goal Position (30)

Set the DYNAMIXEL's goal position through the Goal Position(30). The available position range is 0 to 1,023 (0×3FF) and the per unit value is 0.29 °. If the Goal Position(30) is set over the [CW/CCW Angle Limit\(6, 8\)](#), the Status Packet transmits Angle Limit Error Bit (0×01) via its ERROR field. In the case, providing that Angle Limit Error Bit(0×01) in the [Alarm LED\(17\)/Shutdown\(18\)](#) is set, the Alarm LED will start blinking and the motor's output will be 0 [%].



The picture above is the front view of DYNAMIXEL

NOTE

If it is set to Wheel Mode, Goal Position value is not used.

Moving Speed (32)

It is a moving speed to Goal Position. The range and the unit of the value may vary depending on the operation mode.

- **Joint Mode** 0 ~ 1,023(0x3FF) can be used, and the unit is about 0.111rpm. If it is set to 0, it means the maximum rpm of the motor is used without controlling the speed. If it is 1023, it is about 114rpm. For example, if it is set to 300, it is about 33.3 rpm.

NOTE

Please check the maximum rpm of the DYNAMIXEL. The motor cannot exceed the maximum rpm with the higher Moving Speed value.

- **Wheel Mode** 0 ~ 2,047(0x7FF) can be used, the unit is about 0.1%. If a value in the range of 0 ~ 1,023 is used, it is stopped by setting to 0 while rotating to CCW direction. If a value in the range of 1,024 ~ 2,047 is used, it is stopped by setting to 1,024 while rotating to CW direction. That is, the 10th bit becomes the direction bit to control the direction. In Wheel Mode, only the output control is possible, not speed. For example, if it is set to 512, it means the output is controlled by 50% of the maximum output.

Torque Limit (34)

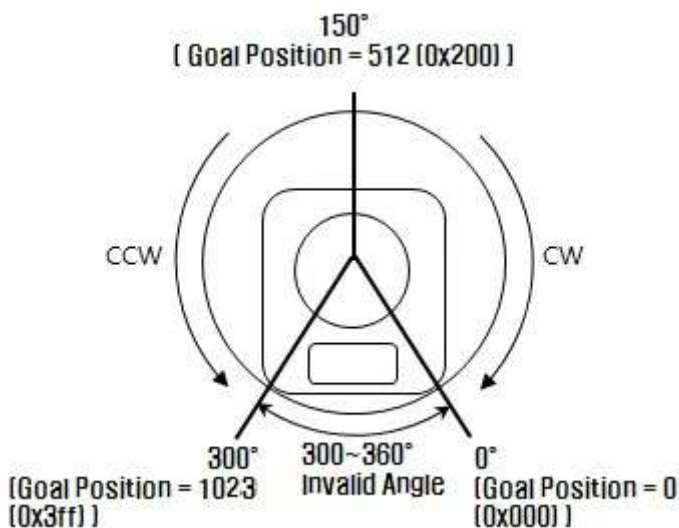
It is the value of the maximum torque limit. 0 ~ 1,023(0×3FF) is available, and the unit is about 0.1%. For example, if the value is 512, it is about 50%; that means only 50% of the maximum torque will be used. When the power is turned on, Torque Limit(34) is reset to the value of [Max Torque\(14\)](#).

NOTE

If the function of [Alarm LED\(17\)/Shutdown\(18\)](#) is triggered, the motor loses its torque because Torque Limit(34) becomes 0. Once error conditions are resolved and Torque Limit(34) is changed to the value other than 0, the motor can be operated again.

Present Position (36)

It is the present position value of DYNAMIXEL. The range of the value is 0~1023 (0×3FF), and the unit is 0.29 [°].



The picture above is the front view of DYNAMIXEL.

WARNING

CAUTION : If it is set to Wheel Mode, the value cannot be used to measure the moving distance and the rotation frequency.

Present Speed (38)

It is the present moving speed. ~~02,047 (0×7FF) can be used. If a value is in the range of 01,023,~~ it means that the motor rotates to the CCW direction. If a value is in the range of 1,024~2,047, it means that the motor rotates to the CW direction. That is, the 10th bit becomes the direction bit

to control the direction, and 0 and 1,024 are equal. The unit of this value varies depending on operation mode.

- Joint Mode

The unit is about 0.111rpm. For example, if it is set to 300, it means that the motor is moving to the CCW direction at a rate of about 33.3rpm.

- Wheel Mode

The unit is about 0.1%. For example, if it is set to 512, it means that the torque is controlled by 50% of the maximum torque to the CCW direction.

Present Load (40)

It means currently applied load. The range of the value is 0~2,047, and the unit is about 0.1%. If the value is 0~1,023, it means the load works to the CCW direction. If the value is 1,024~2,047, it means the load works to the CW direction. That is, the 10th bit becomes the direction bit to control the direction, and 1,024 is equal to 0. For example, the value is 512, it means the load is detected in the direction of CCW about 50% of the maximum torque.

Bit	15 ~ 11	10	9 ~ 0
Value	0	Load Direction	Data (Load Ratio)

NOTE

CCW Load : Load Direction = 0, CW Load : Load Direction = 1

NOTE

Present load is an inferred value based on the internal output value; not a measured value using torque sensor, etc. Therefore, it may be inaccurate for measuring weight or torque. It is recommended to use it for predicting the direction and size of the force being applied to the joint.

Present Voltage (42)

It is the size of the present voltage supplied. This value is 10 times larger than the actual voltage. For example, when 10V is supplied, the data value is 100 (0×64)

For more details, see [Min/Max Voltage Limit \(12, 13\)](#).

Present Temperature (43)

It is the internal temperature of DYNAMIXEL in Celsius. Data value is identical to the actual temperature in Celsius. For example, if the data value is 85 (0×55), the current internal temperature is 85°C.

For more details, see [Temperature Limit \(11\)](#).

Registered Instruction (44)

Indicates whether the Write Instruction is registered by [Reg Write Instruction](#)

Value	Description
0	No instruction registered by REG_WRITE.
1	Instruction registered by REG_WRITE exists.

NOTE

If ACTION instruction is executed, the Registered Instruction (44) will be changed to 0.

Moving (46)

Value	Description
0	Goal position command execution is completed
1	Goal position command execution is in progress

Lock (47)

Value	Description
0	EEPROM area can be modified
1	EEPROM area cannot be modified

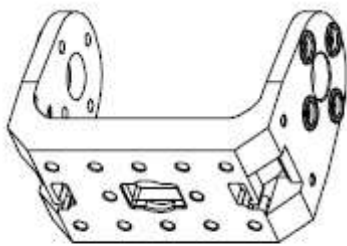
⚠ WARNING

CAUTION : If Lock is set to 1, the power must be turned off and then turned on again to change into 0.

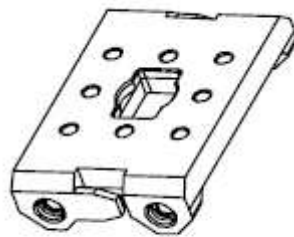
Punch (48)

Minimum current to drive motor. This value ranges from 0×20 to 0×3FF.

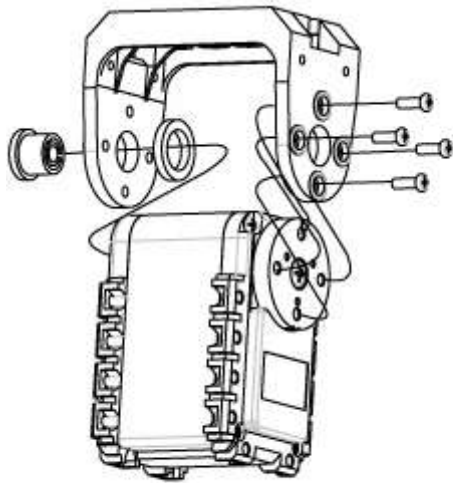
How to Assemble



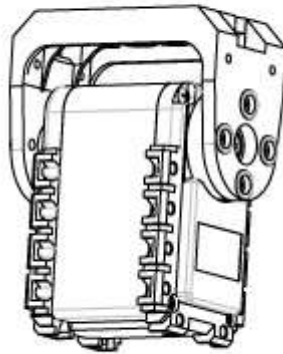
FP04-F2



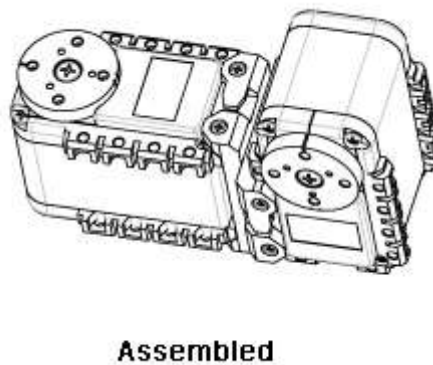
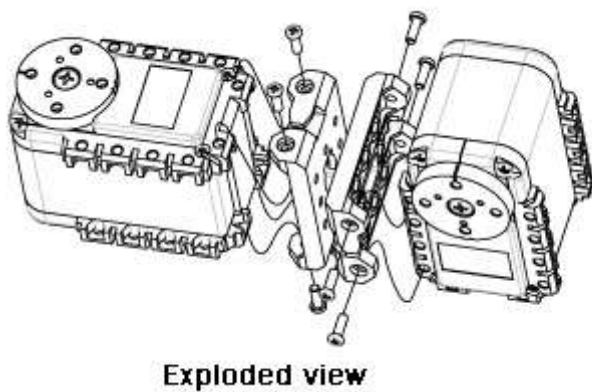
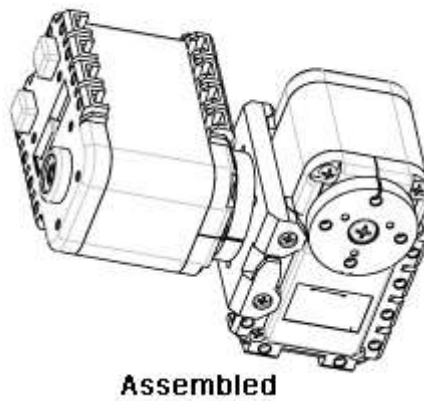
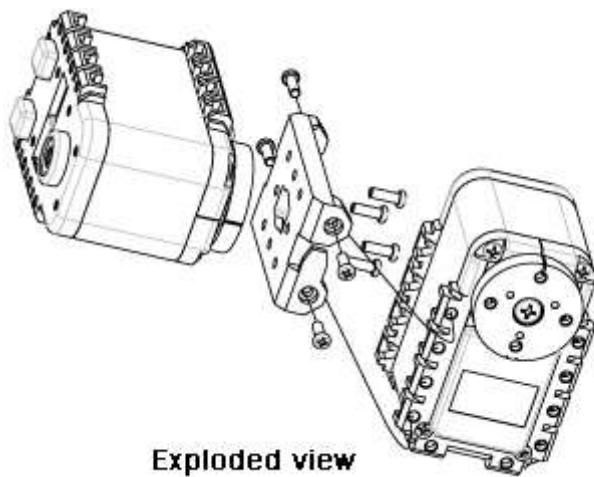
FP04-F3



Exploded view



Assembled



Maintenance

Horn and Bearing Replacement

The horn is installed on the front wheel gear serration of the DYNAMIXEL whereas the bearing set is installed on the back.



Installing the Horn

Place the thrust horn washer into the actuator before inserting the horn. You must carefully align the horn to the wheel gear serration by aligning dots.



Once alignment is properly done, gently push the center of the horn toward the actuator. Make sure that the horn washer is in place as you tighten the bolt.

Installing the Bearing Set

You may need to remove the bearing set from the previous actuator and reinstall it into the new actuator. The bearing set can also be purchased separately. As bearing set is rotating freely, therefore alignment is not required when assembling to DYNAMIXEL.



Reference

i NOTE

NOTE [Compatibility Guide](#) [Harness Compatibility](#)

Certifications

Please inquire us for information regarding unlisted certifications.

FCC**AX-12**** NOTE**

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

** WARNING**

WARNING Any changes or modifications not expressly approved by the manufacturer could void the user's authority to operate the equipment.

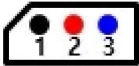
AX-12+ / AX-12A** NOTE**

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

⚠ WARNING

WARNING Any changes or modifications not expressly approved by the manufacturer could void the user's authority to operate the equipment.

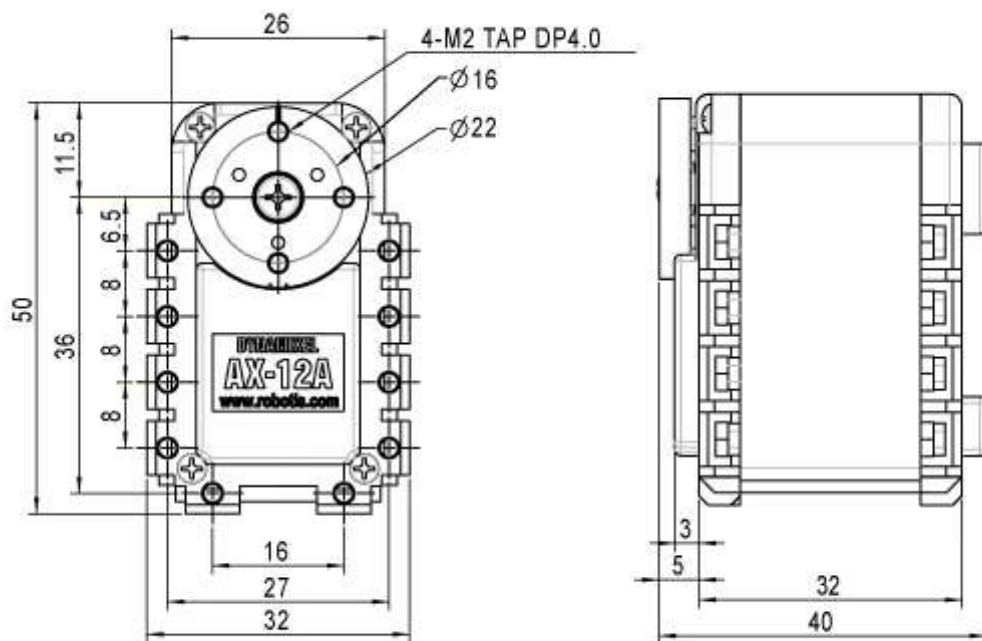
Connector Information

Item	TTL
Pinout	1 GND 2 VDD 3 DATA
Diagram	
Housing	 MOLEX 50-37-5033
PCB Header	 MOLEX 22-03-5035
Crimp Terminal	MOLEX 08-70-1039
Wire Gauge for DYNAMIXEL	21 AWG

🔥 DANGER

WARNING: To enhance user safety and to prevent proprietary risk or damage, **be sure to check the pinout installed on DYNAMIXEL and the board.** The Pinout of DYNAMIXEL may differ depending on a manufacturer of connector.

Drawings

**TIP**

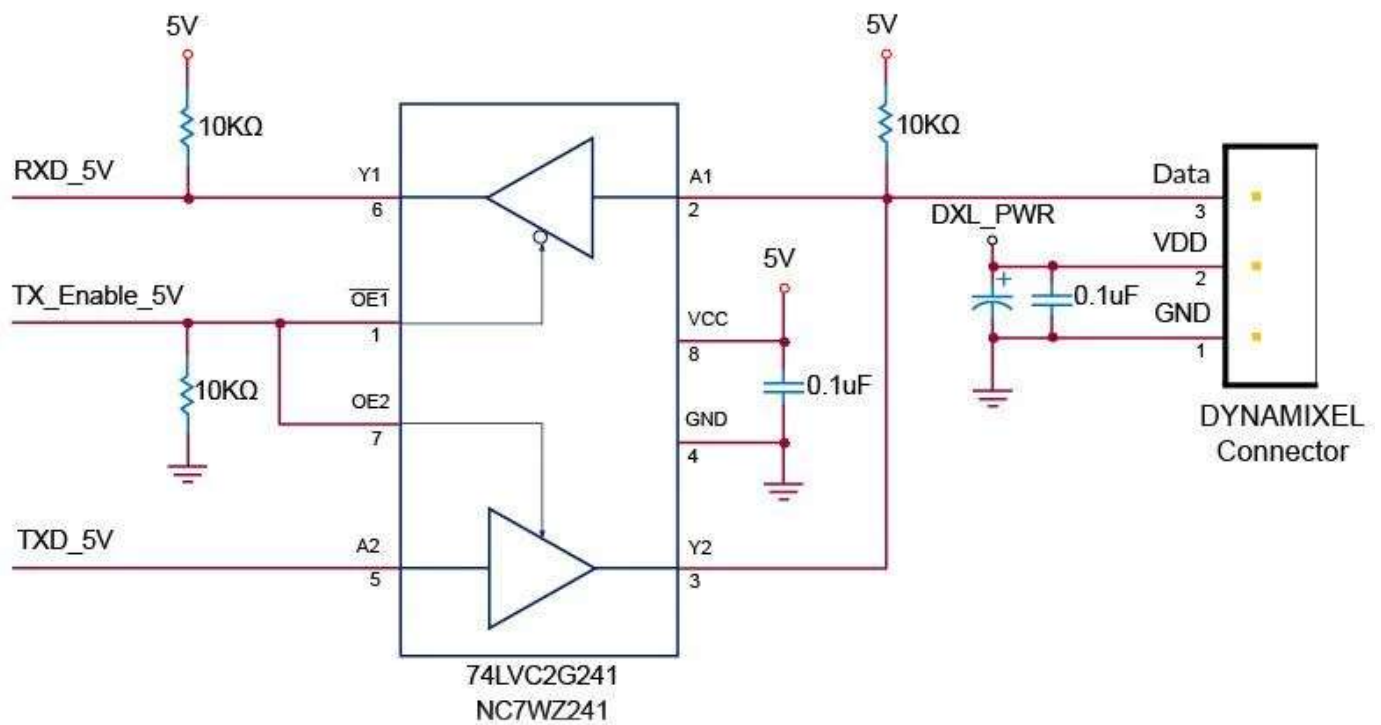
Please also checkout [ROBOTIS Download Center](#) for software applications, 3D/2D CAD, and other useful resources!

Communication Circuit

To control the DYNAMIXEL actuators, the main controller needs to convert its UART signals to the half duplex type. The recommended circuit diagram for this is shown below.

TTL Communication

TTL Communication Circuit



i NOTE

Above circuit is designed for 5V or 5V tolerant MCU. Otherwise, use a Level Shifter to match the voltage of MCU.